Angled and negative work

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Learning target: Use force direction to determine work’s sign and size.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 14.*
15 m. The constant-Δv straight-line model omits full orbital dynamics.

02. Worked example: explain the model
Friction negative; perpendicular force does zero work.

03. Read and interpret the model
Positive, negative, zero respectively.

04. Apply the physics
+24 J,-6 J, net 18 J.

05. Explain your reasoning
Its parallel component is zero, so it does no instantaneous work; can change direction without changing kinetic energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 5.*
24 kg·m/s right, so the vector sum remains zero.

07. Exit check
-20 J.

08. STRETCH: test the model
Net +12 J; doubled friction work -16 J gives net +4 J.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 14.*
$J=7.5\ N\cdot s$; $v_f\approx0.5$ m/s from $mv_i+J=mv_f$.

02. Worked example: explain the model
Friction negative; perpendicular force does zero work.

03. Read and interpret the model
Positive, negative, zero respectively.

04. Apply the physics
Rope 86.6 J; friction -20 J; net 66.6 J.

05. Explain your reasoning
Its parallel component is zero, so it does no instantaneous work; can change direction without changing kinetic energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 5.*
Other velocity -8 m/s; total K=480 J, supplied by released internal energy.

07. Exit check
-10 J; component opposes displacement.

08. STRETCH: test the model
Pull $20\times3\times\cos 60°=30$ J; friction -12 J; sum 18 J. Vertical forces do no work for horizontal displacement.